

To Do or Not To Do? Managing Intentions with Technology

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ABSTRACT

An overarching goal of human-computer interaction is to build technologies that empower users to do what they want to do i.e. to realise their intentions. To achieve high-level goals and values, people have to do certain things and not do other things. Written for the CHI 2023 Doctoral Consortium, this article briefly describes the initial results from two studies which explore how reminders can help people to exhibit and inhibit behaviours, respectively. The first study demonstrates that metacognitive reflection guides reminder setting for everyday intentions. The second study explores how reminders to reflect can empower individuals to use social media more intentionally. These findings are used to inform future directions for research and design.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **HCI theory, concepts and models**.

KEYWORDS

prospective memory, reminders, social media addiction, end-user programming, reflection, metacognition, confidence, ecological methods, in-the-wild

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1 FOREWORD AND PROGRAMME DETAILS

I am delighted to have the opportunity to apply to be a part of CHI Doctoral Consortium 2023. By the time of the conference, I will be 18 months into my PhD at University College London (UCL), with a projected completion date in Autumn 2024. I am completing my PhD as part of the Ecological Study of the Brain Doctoral Training programme (EcoBrain DTP, ecologicalbrain.org). This programme stretches over four years, with the first year encompassing three rotations with different departments, and the final three years focused on doctoral research.

The EcoBrain DTP is an multidisciplinary PhD training programme that recruits students to measure and explore brain processes and behaviour in the real world. My work recognises that

many cognitive processes, such as memory and attention, are increasingly mediated by technology. Subsequently, understanding human-computer interaction has become a critical aspect of the ecological study of the brain. The first year of my PhD has explored the relationship between cognition and technology from many angles, with the overarching research question: **Why do people choose to (or not to) use technologies to support their cognition?** This question has led me to write on the complexity of reminder setting in everyday life (in prep), the privacy implications of assistive technologies [41], new theoretical conceptualisations of ‘creepy’ technologies (under review), and the consequences of perceived machine consciousness in interaction (in press for CHI 2023). As I cannot describe all these various projects within this short doctoral consortium application, I will present my work exploring reminders.

2 INTRODUCTION

Far from a brain-in-a-jar, the brain has been described as extended, distributed, embodied and embedded [35]. In cognitive neuroscience, the use of external sources to reduce information-processing demands is referred to as ‘cognitive offloading’ [35]. Increasingly, people ‘offload’ their cognition using digital technologies. Our ability to remember past events is supported by photographs, videos, audio files, documents, while calendars, reminders, alarms and to-do lists support our memory for future events. The use of reminders (also known as ‘intention offloading’) can hugely increase the accuracy of prospective memory [14, 20–22]. Reminders and other technologies have great potential to improve people’s daily functioning across the general population as well as those who suffer from cognitive impairments [20]. There is an opportunity to leverage these innovations to extend the independence and well-being of an ageing population [10, 36, 41].

An overarching motivation of human-computer interaction is to build technologies that empower users to do what they want to do i.e. realise their intentions. To achieve high-level goals and values, people have to do certain things and *not* do other things. For example, if I want to be more productive, I may want to spend more time writing and less time scrolling on social media. In the two sections of this paper, I present how intention offloading can support both *doing* and *not doing*, respectively. The first section describes a study which builds on previous work into intention offloading, by investigating how people use reminders in everyday life to help them remember what they intend to do. The second section describes a study which explores how a reminder can help people to reduce an undesirable behaviour, namely unintentional use of social media. With this dual focus, this work attempts to cover a wider spectrum of intentions, framed in both the positive (‘to do’) and the negative (‘not to do’). Importantly, both studies look

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at intentions embedded in everyday life, embracing the complexity of the ecological brain and its plans.

3 REMINDERS TO REMEMBER INTENTIONS

Remembering and fulfilling tasks and obligations is a critical executive function, allowing people to live healthy, social, and functional lives. While cognitive processes such as attention, motivation and action are all required to fulfil an intention, prospective memory is required to *remember* delayed intentions. However, prospective memory can be cognitively taxing, relying on either spontaneous recall (which can decline with age [12]) or purposeful rehearsal of intentions and monitoring of passing time. These factors explain why prospective memory failures make up the majority of memory failures in all age groups [9, 31, 43]. More positively, it has been demonstrated that the use of reminders, such as calendars, and alarms, can hugely increase the accuracy of prospective memory, by offloading intentions to an external memory source [14, 17, 21, 22].

One early study that explicitly focused on the use of external memory aids for prospective memory was conducted by Marsh et al. [29]. Participants were asked to list all their upcoming plans for the following week. Seven days later, participants returned and reported what they had actually achieved in this time. Of 135 participants, 57.8% reported using a diary or planner ("recorders"). However, the study did not look to see which individual plans were offloaded, nor what triggered them to offload these plans. In addition, this work was conducted over 20 years ago, so the rate of offloading may have changed significantly with the availability of connected mobile devices.

More recently, Wójcik et al. [44]) adapted the paradigm described by Marsh et al. to explore the differences between time-based or event-based intentions. Time-based intentions must be performed at or by a particular time (e.g. attending a meeting at 2pm), while event-based intentions must be performed when a certain event triggers them (e.g. charging a device when it runs low on battery). For a week, participants submitted their daily intentions. Amongst other measurements, the researchers investigated how the use of memory aids affected the recollection and fulfilment of intentions. They found that the proportion of plans which were offloaded predicted the total proportion of plans which were fulfilled. However, they did not conduct any analysis on the participants' motivations for using these memory aids, nor the type of memory aids that they were using. In our study, we adapt the method presented by Wójcik et al. to investigate the triggers for intention offloading, and the common ways in which people offload their everyday intentions.

Previous research has looked into the triggers of intention offloading, by focusing on the precise moment when someone decides to set a reminder within web-based tasks [19, 20, 35, 38]. In these tasks, participants have to recall where to drag a circle over a few seconds. They can 'offload' this obligation, by dragging the circle towards its destination ahead of time; this is analogous to leaving your keys by the front door, and so remember to take them as you leave. These experiments have found that an individual's unaided confidence in the task can predict their use of reminders [17, 19]. These findings have been formalised into the metacognitive account of cognitive offloading, which hypothesises that if someone has low confidence in their ability to remember an intention, they are

more likely to offload the intention to an external memory source, such as a calendar or an alarm. However, the effect of confidence on reminder setting has not yet been demonstrated in everyday life, where intentions can be more complex and held over longer periods of time. Hence, the first two hypotheses for our study were as follows:

- **H1: Plans for which a reminder has been set will be associated with lower confidence than plans without a reminder.**
- **H2: Participants with lower confidence in their unaided memory will set reminders for a larger proportion of their plans.**

3.1 Participants

We recruited 112 participants via the research platform Prolific. The gender split was 52 female, 59 male, and 1 preferring to self describe. The mean age was 35 years old (min: 19, max: 70). Participants were required to be fluent in English. Over half of the participants (63) returned to the second survey link four or five days later.

3.2 Method

The survey was created using Qualtrics XM software. As well as asking about the participant demographics, we also asked participants to list all their upcoming plans for the following three days (not including the day of the first survey). The question was accompanied by a further instructive graphic shown in Figure 1.

Up to eight of the plans submitted by participants were randomly sampled for further questions. Among other questions, we asked whether they had set a reminder for this plan, and their likelihood of remembering the plan both with and without the reminder i.e. their confidence. They were asked to return to a second survey four to five days later.

In the second survey, the participants were asked whether they completed each plan or not. If they answered 'Yes' to item 2, this plan was categorised as 'fulfilled', and the survey looped straight to the next plan. If the participant answered 'Yes, but not at the time I originally intended' or 'No', the plan was categorised as 'not fulfilled'. For non-fulfilled plans, participants were asked why they didn't fulfil the plan. Some plans may have been remembered, but not fulfilled for other reasons.

3.3 Analysis

All analyses were conducted in R using RStudio. For comparisons between offloaded and non-offloaded plans, we only included participants who had reported at least one of each. To test hypothesis 1a and 2, we conducted paired T-tests. For hypothesis 1b, we conducted a Pearson correlation analysis.

3.4 Results

Participants submitted between 3 and 30 plans, with an average of 8 plans. In total, our 112 participants submitted 926 plans, of which 719 were subject to further questions.

Out of 112 participants, 71% reported using a calendar or diary, and 60% reported using to-do lists more than once a week. On average, participants set a reminder for 34% of the randomly sampled plans. Digital reminders were the most commonly used, set for 137



How to recall all your plans for the next three days

Please do NOT list:

- Routine activities that you do automatically everyday e.g. having dinner, or brushing your teeth
- Duplicate plans

Example plans:

- Meet John
- Complete presentation
- Go to cinema
- Catch train to Manchester
- Finish book
- Buy washing-up liquid
- Bake cake
- Call Mum

1. List all the plans that come to your mind immediately. 

2. If you use a calendar or diary, refer to that.  

3. If you use to-do lists (or something else), refer to those.  

4. Think through and list any of the following:

- a. Household chores and tasks 
- b. Upcoming events or dates with friends and family 
- c. Preparations or assignments for work or school 
- d. Upcoming non-routine journeys 
- e. Shopping trips or specific items that you need or purchase or book 
- f. Emails, messages, or packages you need to send 
- g. Appointments or phone calls you need to make  
- h. Health-related activities, such as taking medications or exercising 

Figure 1: An instructive graphic describing how participants could recollect and describe their plans for the three days following the day of initial participation.

plans, followed by physical reminders (114) and social reminders (76).

We found that the average unaided confidence for offloaded plans were rated 67%, and the average unaided confidence for nonoffloaded plans was 85%. This difference was statistically significant, $t(78) = 7.21, p < 0.01$. We also found that our participants' average confidence across all plans was negatively correlated with the overall proportion of plans they offloaded, $r = 0.44, p < 0.001$. Hence, both hypotheses were supported. These relationships have been found previously in artificial experimental tasks, but here it is demonstrated for the first time in naturally occurring intentions.

Half of the participants (63) returned to the second survey, and reported on the 414 plans they submitted. Of these plans, 281 (68%) were fulfilled as planned. The remaining 133 plans (32%) were either not fulfilled, or fulfilled at a different time (i.e. later). Of the 133 plans which were not fulfilled, 45 (33%) were reprioritised, 28 (21%) were barely possible to perform due to circumstances, 24 (18%) were delayed as the participants was waiting for a better opportunity, 20 (15%) were forgotten (or necessary information was forgotten, and 14 (10%) were no longer necessary to complete.

Of the 414 plans sampled in the second survey, 143 (35%) had a reminder set for them originally. Of these, 104 were fulfilled (73%). 39 offloaded plans (27%) were not fulfilled, eight of which were due to forgetting. For the 271 plans for which no reminder was set, 177 (65%) were fulfilled. 94 non-offloaded plans (35%) were not fulfilled, of which 12 were due to forgetting. Note that each participant could submit multiple plans, so these plans are not statistically independent. Future work will validate the efficacy of reminders on fulfilment and forgetting using path analysis with mixed-effect linear regression, to account for hierarchical data structure [3, 18].

3.5 Discussion and Research Propositions

Our results show that people already make extensive use of reminders to help them remember future intentions, more so than in 1998 when Marsh et al. [29] conducted their research. The findings also suggest that digital reminders are the most common way that people offload their plans, with digital reminders involved in 54% of all offloaded plans. However, our participants were recruited via an online survey platform, so they may be more comfortable with digital tools than the general population and have distinct

plans and schedules. Future work should assess the reminder use of harder to reach populations.

Our results are the first to provide evidence for the metacognitive account of intention offloading in real life plans [11, 35], suggesting that confidence is a significant parameter of intention offloading outside of experimental web-tasks. This supports the intuitive proposition that people rely on external sources of memory when they doubt the capabilities of their own unaided memory. Some people will be over-confident (or under-confident) leading to sub-optimal use of reminders. Therefore, we plan to develop an intervention which trains metacognitive skills, leading to more accurate assessment of confidence and subsequently more optimal use of reminders. The next step is to implement this training protocol within people's everyday lives, without invading their privacy.

Metacognitive training may be able to get people to set more reminders when they need them. However, our data suggest that reminders are not always successful in prompting the recall and fulfilment of future plans; 32% of offloaded plans were not fulfilled as expected. Reminders may be missed or ignored, and could benefit from being made more salient and efficient. We are currently working to explore how augmented reality could be used to facilitate immediately accessible and relevant reminders in high memory load settings, such as hospitals and care homes. We are conducting interviews with nurses to capture how they currently use reminders, and what they most commonly forget.

Recollection and fulfilment of plans can also be enhanced by 'future thinking', or thinking about future plans in rich detail [2, 13]. However, it has not yet been explored if new technologies such as virtual reality (VR) can help the visualisation processes involved in future thinking. Hence, I am currently working to design a VR experience which can help people visualise and formalise their plans for future intentions, and see how this effects their recall and fulfilment. The first step is to hold co-design sessions with people who wish to improve their planning skills, and those who have calendar synaesthesia, so already associate times or days of the week with visual or spatial aspects [34]. Key challenges will be determining the best time scale to address (e.g. planning a single event versus a weekly schedule), and representing tasks and sub-tasks across time and space [1].

4 REMINDERS TO REFLECT ON INTENTIONS

As the first study demonstrates, many people struggle to make their behaviour match their intentions. Alongside the difficulties in exhibiting more desirable actions and behaviours, the inability to *inhibit* unwanted, inappropriate or dangerous behaviours can also contribute to this gap. A modern example of this phenomena is 'digital addiction', where people spend more time than they would like using certain digital technologies. Increasingly people are trying to improve their digital wellbeing through 'digital detox' [33], 'digital self-control' [5, 40] and 'technology unuse' [25] strategies. People use diverse strategies to change their digital behaviour, including physical actions, such as leaving a device elsewhere or turning the device off for periods of time. There are also 'digital wellbeing apps' which are specifically designed for promoting more conscious use of smartphones [30], including screen-time caps and lock-out tasks [24].

Despite this range of interventions, previous work has shown that people often fail to maintain their digital detox. Schoenebeck [39] found that in a cohort of people giving up Twitter for lent, a 40 day period, at least 26% of them returned before that time was up. Research has demonstrated users' frustration with both highly-restricting lock-out apps as well as those that are easy to avoid [23]. These findings suggest that while offloading self-control to restrictive interventions can reduce usage to some degree, users still maintain a desire to use the device or app, and may circumvent interventions to do so.

In this section, I report a study which focused on building an intervention to reduce unintentional use of Instagram. This study was conducted alongside Elena Radziunaite, as part of her MSc thesis project. While problematic use has been documented across many social network sites (SNS) [6, 32, 37], we chose to focus on Instagram use, as this is one of the most popular and heavily used SNS globally [16]. Instagram uses a variety of addictive design features [4, 27]. We are aware of one study which explored the effect of a short break from Instagram on users' wellbeing [15], but did not discuss how digital interventions could assist or affect this process.

Our intervention asked the following question as users opened the Instagram app: "What brought you to Instagram?" This question does not require additional context awareness, as it asks users to decide what intentional usage looks like to them, rather than adopt a preconceived notion from the researcher. The open-ended question prompts instantaneous reflection at the moment of opening the app, inviting our participants to critically observe their Instagram usage [7, 8, 42]. Hence, this intervention can adapt to the volatile intentions of users, while supporting their higher-level intention to not use Instagram mindlessly [28]. To evaluate this intervention, we used a qualitative approach guided by research questions rather than hypotheses. The findings were extracted through qualitative analysis of interview data, rather than statistical analyses. Our research questions were as follows:

- **RQ1: Can a prompt for reflection increase participants' awareness of their own Instagram use?**
- **RQ2: Can a prompt for reflection help participants use Instagram more intentionally?**

4.1 Participants

For the intervention study, we recruited 17 iPhone users between the ages of 23 and 26. Seven participants were recruited from an initial interview study (not reported in this paper), and further participants were recruited through Instagram.

4.2 Method

The intervention was assembled by the participants themselves through the Automations functionality on their personal iPhones. Automations are found within the 'Shortcuts' application, which offers a visual scripting tool that allows users to create their own rules and conditions for executing specific tasks on their device. In this case, participants created an automated reminder that popped up whenever they opened Instagram, asking them: 'What brings you to Instagram?' (Figure 2).

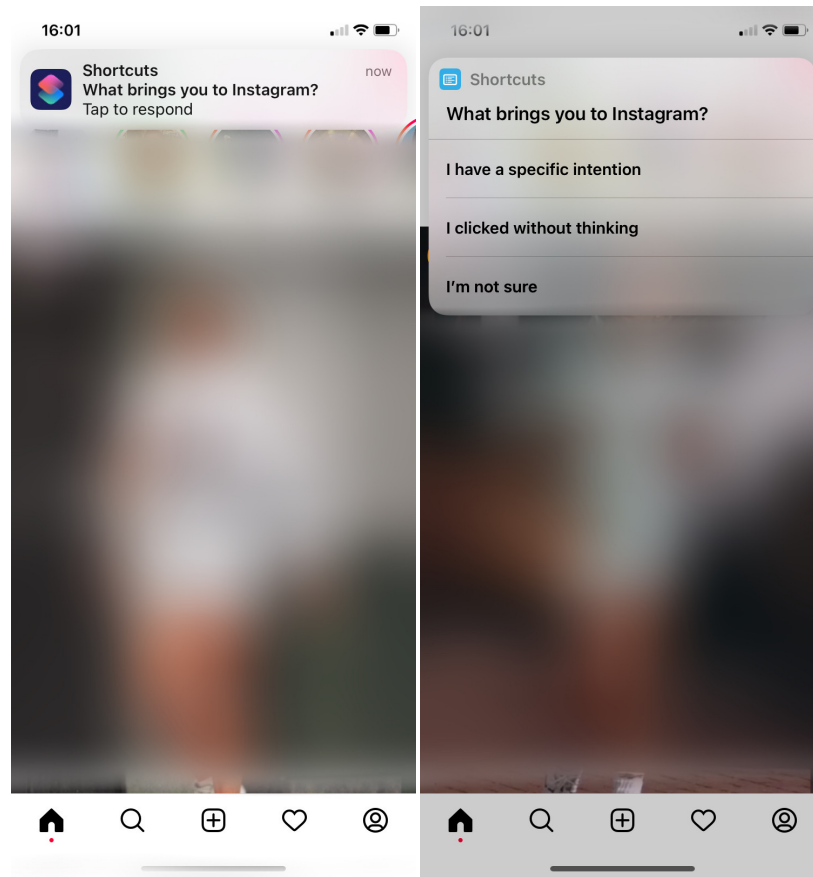


Figure 2: Two screenshots demonstrating how the intervention appears as a user opens Instagram. They are first shown a pop-up with the prompt to reflect on their intention. If they tap to respond, they are shown three options to choose from.

The participants ran the intervention for between five and nine days. This variation in study duration was due to technical difficulties; within the study period, some participants found that the intervention did not work as expected all the time. Semi-structured interviews with 14 of the participants were conducted after the study period. Questions asked during the retrospective interview included:

- What is your overall impression of the intervention?
- How would you change or improve the intervention?
- How could you see yourself using Shortcuts again?

All interview recordings were transcribed verbatim and analysed using affinity diagramming [26]. We identified six key themes.

4.3 Findings and discussion

Six key themes were extracted from the interview data:

- i) varying understandings of ‘intentional usage’
- ii) the wide range of contextual factors that affected these understandings
- iii) social exposure as a design resource for interventions
- iv) reflection as a design resource for interventions

- v) proposed changes to the intervention suggested by the participants
- vi) discovering iOS Automations

The findings give evidence that a reflection-based intervention can meaningfully confront users on their habitual Instagram usage. Rather than controlling the absolute time they spend on the app, this intervention engages users with a prompt for reflection, potentially building their intrinsic self-awareness and self-discipline [8]. Even when the pop-up wasn’t working as expected, some participants found themselves automatically considering their intentions when opening Instagram, suggesting that the pop-up started to develop regular self-evaluative habits in participants. There is also evidence that building the intervention themselves led to psychological investment in its success, and a sense of self-efficacy and accomplishment.

Moreover, our findings resonate with previous work describing that users describe very different intentions for their SNS usage - and for digital wellbeing interventions [7]. The participants were eager to propose design ideas for how those individual needs could be met, such as the use of social exposure, redirection to other applications, and the setting of count-down timers to limit use. We found that iOS Automations provide a relatively easy-to-master

toolkit for the users to implement such ideas. They demonstrated notable enthusiasm for doing so, to manage the time they spent on Instagram and other applications, such as TikTok.

5 FUTURE PLANS

As well as the three directions described in the first section of this report, I will continue exploring the potential for digital wellbeing interventions designed by the end-user. This work will look at how individuals can iterate their own intervention designs to respond to habituation and frustration. Through this two-pronged approach, I hope to build a comprehensive understanding of how technology can be used to help us do more of what we want to do, and less of what we don't.

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